

# Retrieve fraction methods

Use the scaffold, show each step and ask for an adult check when marked.

Name: \_\_\_\_\_ Date: \_\_\_\_\_

1. Read the model: Calculate  $\frac{5}{6} + \frac{7}{12}$ . Copy or complete the first step.

2. Find 37.5% of 640.

3. Simplify 45:60 and find an equivalent ratio with first part 12.

4. Miro says: Miro chooses a familiar method before identifying the number type and relationship. Is this correct? Explain.

# Teacher answers and checking prompts

1. Read the model: Calculate  $\frac{5}{6} + \frac{7}{12}$ . Copy or complete the first step.  
**Answer**  $1 \frac{5}{12}$ .
2. Find 37.5% of 640.  
**Answer** 240.
3. Simplify 45:60 and find an equivalent ratio with first part 12.  
**Answer** 3:4 and 12:16.
4. Miro says: Miro chooses a familiar method before identifying the number type and relationship. Is this correct? Explain.  
**Answer** Name the domain first, represent the relationship, then choose and check the calculation.

# Retrieve fraction methods

Work independently. Show a complete method and check at least one answer.

Name: \_\_\_\_\_ Date: \_\_\_\_\_

1. Calculate  $\frac{5}{6} + \frac{7}{12}$ .

6. Check this result using an inverse, estimate or known fact: Simplify 45:60 and find an equivalent ratio with first part 12.

2. Find 37.5% of 640.

7. Prove or explain your reasoning: A recipe uses  $\frac{3}{4}$  kg for 6 people. How much is needed for 14 people?

3. Simplify 45:60 and find an equivalent ratio with first part 12.

8. Identify and correct this misconception: Miro chooses a familiar method before identifying the number type and relationship.

4. Solve this independently and show every stage: Calculate  $\frac{5}{6} + \frac{7}{12}$ .

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Find 37.5% of 640.

10. Write and solve a similar question to: Simplify 45:60 and find an equivalent ratio with first part 12.

# Teacher answers and checking prompts

1. Calculate  $5/6 + 7/12$ .  
**Answer**  $1\frac{5}{12}$ .
2. Find 37.5% of 640.  
**Answer** 240.
3. Simplify 45:60 and find an equivalent ratio with first part 12.  
**Answer** 3:4 and 12:16.
4. Solve this independently and show every stage: Calculate  $5/6 + 7/12$ .  
**Answer**  $1\frac{5}{12}$ .
5. Use a second method or representation for: Find 37.5% of 640.  
**Answer** Expected result: 240. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Simplify 45:60 and find an equivalent ratio with first part 12.  
**Answer** Expected result: 3:4 and 12:16. A valid check must be shown.
7. Prove or explain your reasoning: A recipe uses  $3/4$  kg for 6 people. How much is needed for 14 people?  
**Answer**  $1\frac{3}{4}$  kg. The explanation must show why the method works.
8. Identify and correct this misconception: Miro chooses a familiar method before identifying the number type and relationship.  
**Answer** Name the domain first, represent the relationship, then choose and check the calculation.
9. Write a complete sentence explaining how you checked one answer.  
**Answer** Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Simplify 45:60 and find an equivalent ratio with first part 12.  
**Answer** Answers vary. The new question must use the same mathematical structure and include a correct solution.

# Retrieve fraction methods

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: \_\_\_\_\_ Date: \_\_\_\_\_

1. Prove or explain your reasoning: A recipe uses  $\frac{3}{4}$  kg for 6 people. How much is needed for 14 people?

6. Create a new example that tests the same idea as: Simplify 45:60 and find an equivalent ratio with first part 12.

2. Prove the answer to this question, rather than only calculating it: Calculate  $\frac{5}{6} + \frac{7}{12}$ .

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Find 37.5% of 640.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 3:4 and 12:16.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro chooses a familiar method before identifying the number type and relationship.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

# Teacher answers and checking prompts

1. Prove or explain your reasoning: A recipe uses  $\frac{3}{4}$  kg for 6 people. How much is needed for 14 people?  
**Answer**  $1\frac{3}{4}$  kg. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: Calculate  $\frac{5}{6} + \frac{7}{12}$ .  
**Answer** Expected result:  $1\frac{5}{12}$ . Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Find 37.5% of 640.  
**Answer** Expected result: 240. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 3:4 and 12:16.  
**Answer** One valid reconstruction is: Simplify 45:60 and find an equivalent ratio with first part 12. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro chooses a familiar method before identifying the number type and relationship.  
**Answer** Name the domain first, represent the relationship, then choose and check the calculation.
6. Create a new example that tests the same idea as: Simplify 45:60 and find an equivalent ratio with first part 12.  
**Answer** Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.  
**Answer** Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.  
**Answer** Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.  
**Answer** Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.  
**Answer** Answers vary. The explanation must distinguish the mathematical structure from the changed value.